

December 2020 (MODULE-1)

1. Determine the rank of the matrix, $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -2 \\ 2 & 4 & -3 \end{bmatrix}$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -2 \\ 0 & 3 & -2 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of $A = \underline{\underline{2}}$

11) a. Solve the following linear system of equations using Gauss elimination method

$$\begin{aligned} x+y+z &= 6 \\ x+2y-3z &= -4 \\ -x-4y+9z &= 18 \end{aligned}$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ -1 & -4 & 9 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} 6 \\ -4 \\ 18 \end{bmatrix}$$

$$A:B = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & -3 & -4 \\ -1 & -4 & 9 & 18 \end{array} \right] \quad \begin{aligned} R_2 &\rightarrow R_2 - R_1 \\ R_3 &\rightarrow R_3 + R_1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -4 & -10 \\ 0 & -3 & 10 & 24 \end{array} \right] \quad R_3 \rightarrow R_3 + 3R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -4 & -10 \\ 0 & 0 & -2 & -6 \end{array} \right] \quad \begin{aligned} \rho(A:B) &= \rho(A) = n = 3. \\ \text{So, unique solution.} \end{aligned}$$

$$\begin{aligned} x+y+z &= 6 \\ y-4z &= -10 \\ -2z &= -6 \Rightarrow \underline{z=3} \end{aligned}$$

$$y - 12 = -10 \Rightarrow y = -10 + 12 = \underline{2}$$

$$x + 2 + 3 = 6 \Rightarrow \underline{x=1}$$

$$X = \underline{\underline{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}}$$

b) Find eigen values and eigen vectors of the matrix.

$$A = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

So, the Characteristic Equation is,
 $\lambda^3 - \text{tr}(A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$
 $\lambda^3 - 6\lambda^2 + (5+3+3)\lambda - 6 = 0$
 $\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\therefore$ Eigen values are, $\lambda = 1, 3, 2$

To find Eigen vector corresponding to $\lambda = 1$, Put $\lambda = 1$ in Eq ①.

$$\begin{bmatrix} 1-\lambda & 1 & 2 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & 3-\lambda \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_2 \leftrightarrow R_1$$

$$\sim \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2 \quad \sim \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \rho(A) = 2 \quad n = 3 \quad \therefore n - r = 3 - 2 = \underline{\underline{1}}$$

$$-x_1 + x_2 + x_3 = 0$$

$$x_2 + 2x_3 = 0$$

$$\text{Put } x_3 = k \Rightarrow x_2 + 2k = 0 \Rightarrow x_2 = \underline{\underline{-2k}}$$

$$-x_1 - 2k + k = 0 \Rightarrow -x_1 = k \Rightarrow x_1 = -k$$

$$X = \begin{bmatrix} -k \\ -2k \\ k \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} k = 1$$

To find Eigen vector corresponding to $\lambda = 3$, Put $\lambda = 3$ in Eq ①.

$$A_2 = \begin{bmatrix} -2 & 1 & 2 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_2 \rightarrow 2R_2 - R_1$$

$$\sim \begin{bmatrix} -2 & 1 & 2 \\ 0 & -3 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad R_3 \rightarrow 3R_3 + R_2 \quad \sim \begin{bmatrix} -2 & 1 & 2 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \rho(A) = 2 \quad n = 3$$

$\therefore n - r = 1$

$$-2x_1 + x_2 + 2x_3 = 0$$

$$-3x_2 = 0 \Rightarrow x_2 = \underline{0}$$

$$-2x_1 + 2x_3 = 0$$

$$\text{Put } x_3 = k \Rightarrow -2x_1 = -2k$$

$$x_1 = \underline{k}$$

$$X_2 = \begin{bmatrix} k \\ 0 \\ k \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}_{k=1}$$

To find eigen vector corresponding to $\lambda = 2$, Put $\lambda = 2$ in eq ①.

$$A_3 = \begin{bmatrix} -1 & 1 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_2 \rightarrow R_2 - R_1$$

$$\sim \begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \quad R_3 \rightarrow R_3 + R_2 \quad \sim \begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$P(A) = 2, \quad n = 3, \quad n - r = \underline{1}$$

$$-x_1 + x_2 + 2x_3 = 0$$

$$-x_2 - x_3 = 0$$

$$\text{Put } x_2 = k$$

$$-k = x_3$$

$$-x_1 + k - 2k = 0 \Rightarrow -x_1 = k$$

$$x_1 = \underline{-k}$$

$$X_3 = \begin{bmatrix} -k \\ k \\ -k \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}_{k=1} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

2. Show that the Quadratic form $4x^2 + 12xy + 13y^2$ is Positive definite.

$$A = \begin{bmatrix} 4 & 6 \\ 6 & 13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

So, the characteristic equation is,

$$\lambda^2 - \text{tr}(A)\lambda + |A| = 0$$

$$\lambda^2 - 17\lambda + 16 = 0$$

$\therefore$ Values of $\lambda = 16, 1$

all values are positive.

$\therefore$ the Quadratic form is positive definite.

12 a. Show that the equation,

$$x+y+z=a, \quad 3x+4y+5z=b, \quad 2x+3y+4z=c$$

(i) have no solution, if $a=b=c=1$

(ii) have many solutions if $a=b/2=c=1$

Ans:- $A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 4 & 5 \\ 2 & 3 & 4 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $B = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

$$A:B = \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 3 & 4 & 5 & b \\ 2 & 3 & 4 & c \end{array} \right] \quad \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b-3a \\ 0 & 1 & 2 & c-2a \end{array} \right] \quad R_3 \rightarrow R_3 - R_2 \quad \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b-3a \\ 0 & 0 & 0 & a-b+c \end{array} \right]$$

(i) $a=b=c=1$

$$(A:B) \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad \begin{array}{l} \rho(A:B)=3 \quad \rho(A)=2 \\ \therefore \rho(A:B) \neq \rho(A) \\ \text{no solution} \end{array}$$

(ii) $a=b/2=c=1$ or $a=1, b=2, c=1$

$$(A:B) \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \rho(A)=2 < n=3$$

b) Find the matrix of transformation that diagonalize the matrix.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Ans - So the characteristic equation is,

$$\lambda^3 - \text{tr}(A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$$

$$\lambda^3 - 12\lambda^2 + (8 + 14 + 14)\lambda - 32 = 0$$

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$\therefore \lambda = 8, 2, 2$$

$$\text{Consider, } (A - \lambda I)x \Rightarrow \begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix}$$

To find eigen vector corresponding to $\lambda = 8$, Put $\lambda = 8$ in eq ①.

$$A_1 = \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 + R_1 \end{array}$$

$$A_1 = \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} \quad \rho(A) = 2, n = 3, n - \rho = \underline{1}$$

$$-2x_1 - 2x_2 + 2x_3 = 0$$

$$-3x_2 - 3x_3 = 0 \quad \text{Put } x_2 = \underline{k}$$

$$-3k = 3x_3 \Rightarrow x_3 = \underline{-k}$$

$$-2x_1 - 2k - 2k = 0 \Rightarrow -2x_1 = 4k \Rightarrow x_1 = \underline{-2k}$$

$$X_1 = \begin{bmatrix} -2k \\ k \\ -k \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix} k = 1 = \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}$$

To find eigen vector corresponding to $\lambda = 2$, Put $\lambda = 2$ in eq ①

$$A_2 = \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow 2R_2 + R_1 \\ R_3 \rightarrow 2R_3 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 4 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \rho(A) = 1, \quad n = 3, \quad n - \rho = \underline{\underline{2}}$$

$$4x_1 - 2x_2 + 2x_3 = 0$$

$$\text{Put } x_2 = k_1 \text{ \& } x_3 = k_2$$

$$4x_1 - 2k_1 + 2k_2 = 0$$

$$4x_1 = 2k_1 - 2k_2$$

$$x_1 = \frac{k_1}{2} - \frac{k_2}{2} = \frac{k_1 - k_2}{2}$$

$$X_2 = \begin{bmatrix} \frac{k_1 - k_2}{2} \\ k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} \text{ \& } \begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} -2 & 1/2 & -1/2 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$X^{-1}AX = D$$

$$\begin{bmatrix} -2 & 1/2 & -1/2 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} -2 & 1/2 & -1/2 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \underline{\underline{D}}$$



3

If $z = \sin(y^2 - 4x)$. Find the rate of change of z with respect to x at the point $(3, 1)$ with y held fixed.

$$\begin{aligned}\frac{dz}{dx} &= \cos(y^2 - 4x) \times -4 \\ &= -4 \cos(y^2 - 4x)\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial x} &= -4 \cos(1^2 - (4 \times 3)) \\ (3, 1) &= -4 \cos(11)\end{aligned}$$

4

Find $\frac{dz}{dt}$ by chain rule, where $z = 3x^2y^2$,
 $x = t^4$, $y = t^3$

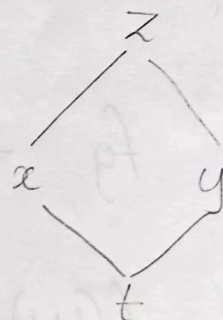
$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \times \frac{dx}{dt} + \frac{\partial z}{\partial y} \times \frac{dy}{dt}$$

$$= 6xy^2 \times 4t^3 + 6x^2y \times 3t^2$$

$$= 6 \times t^4 \times t^6 \times 4t^3 + 6 \times t^8 \times t^3 \times 3t^2$$

$$= 24t^{13} + 18t^{13}$$

$$= \underline{\underline{42t^{13}}}$$



13 a) Find the local linear approximation of

$$\frac{4y}{x+z} \text{ at } (1, 1, 1)$$

b) Find the absolute extrema of the fn
 $f(x, y) = x^2 - 3y^2 - 2x + 6y$ over the square region
with vertices $(0, 0)$ $(0, 2)$ $(2, 2)$ and $(2, 0)$

$$a) \quad L(x, y, z) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0) \\ \times (y - y_0) + f_z(x_0, y_0, z_0)(z - z_0)$$

$$f = \frac{4y}{x+z}$$

$$f_x = \frac{-4y}{(x+z)^2}$$

$$f_y = \frac{4}{x+z}$$

$$f_x(1, 1, 1) = \frac{-4 \times 1}{2^2}$$

$$= \underline{\underline{-1}}$$

$$f_y(1, 1, 1) = 2$$

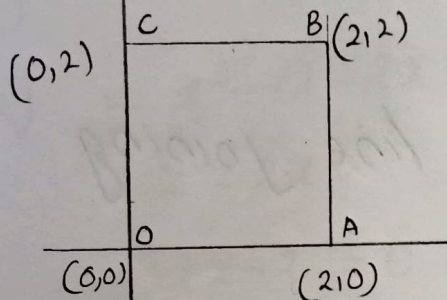
$$f_z = \frac{-4y}{(x+z)^2}$$

$$f_z(1, 1, 1) = -1$$

$$\begin{aligned} L(x, y, z) &= 2 + (-1)(x-1) + 2(y-1) + -1(z-1) \\ &= 2 - x + 1 + 2y - 2 - z + 1 \end{aligned}$$

$$= 2 - x + 2y - z$$

b)



The critical points of $f(x, y)$ in the interior of R are obtained by solving $f_x = 0$, $f_y = 0$

$$f_x = 2x - 2 = 0$$

$$\Rightarrow \begin{aligned} 2x &= 2 \\ x &= 1 \end{aligned}$$

$$f_y = -6y + 6 = 0$$

$$\Rightarrow \begin{aligned} 6y &= 6 \\ y &= 1 \end{aligned}$$

$\therefore$ Critical point is $(1, 1)$

Consider the boundary of the region. It is the combination of 4 line segments.

line joining $(0,0)$ and $(2,0)$

$$y = 0$$

$$f(x,y) = x^2 - 2x$$

$$f' = 2x - 2 = 0$$

$$\Rightarrow 2x = 2$$

$$x = 1$$

The critical point on the line joining $(0,0)$ $(2,0)$ is $(1,0)$

$\therefore$ absolute extrema can occur on the boundary points $(0,0)$ $(2,0)$ and the critical point $(1,0)$

line joining $(2,0)$ and $(2,2)$

$$x = 2$$

$$f(x,y) = f(2,y)$$

$$= 4 - 3y^2 - 4 + 6y$$

$$= -3y^2 + 6y$$

$$f' = -6y + 6 = 0$$

$$\Rightarrow y = 1$$

$\therefore$ the critical point on this line is $(2,1)$

$\therefore$ abs. extrema can occur on the boundary points $(2,0)$ $(2,2)$ and the critical point $(2,1)$

line joining $(2,2)$ $(0,2)$

$$y = 2$$

$$\begin{aligned} f(x,2) &= x^2 - 12 - 2x + 12 \\ &= x^2 - 2x \end{aligned}$$

$$f' = 2x - 2 = 0$$

$$\Rightarrow x = 1$$

$\therefore$ The critical point on this line is

$$(1, 2)$$

$\therefore$ abs. extrema can occur on the boundary points $(2,2)$ $(0,2)$ and the critical point $(1,2)$

line joining $(0,2)$ and $(0,0)$

$$x = 0$$

$$f(0,y) = -3y^2 + 6y$$

$$f' = -6y + 6 = 0$$

$$\Rightarrow y = 1$$

$\therefore$ critical point is $(0,1)$

$\therefore$ abs. extrema can occur on $(0,2)$ $(0,0)$ and the critical point $(0,1)$

(x,y)	$(0,0)$	$(2,0)$	$(2,2)$	$(0,2)$	$(1,0)$	$(2,1)$	$(1,1)$	$(1,2)$	$(0,1)$
$f(x,y)$	0	0	0	0	-1	3	2	-1	3

$\therefore$ abs. minimum value is -1

abs. maximum value is 3.

12. a) If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ find the value of x

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$$

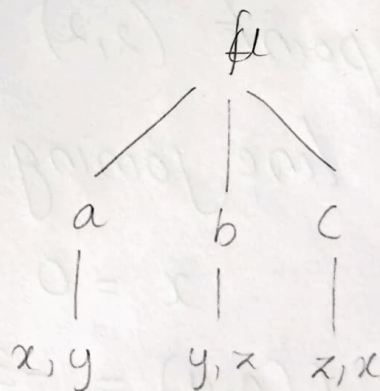
b) Locate all relative extrema of $f(x,y)$
 $= 2xy - x^3 - y^2$

a) Let $a = \frac{x}{y}$, $b = \frac{y}{z}$, $c = \frac{z}{x}$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial a} \times \frac{\partial a}{\partial x} + \frac{\partial u}{\partial c} \times \frac{\partial c}{\partial x}$$

$$= \frac{\partial u}{\partial a} \times \frac{1}{y} + \frac{\partial u}{\partial c} \times \frac{-z}{x^2}$$

$$= \frac{\partial u}{\partial a} \times \frac{1}{y} - \frac{\partial u}{\partial c} \times \frac{z}{x^2} \longrightarrow \textcircled{1}$$



$$\begin{aligned}
 \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial a} \times \frac{\partial a}{\partial y} + \frac{\partial u}{\partial b} \times \frac{\partial b}{\partial y} \\
 &= \frac{\partial u}{\partial a} \times \frac{-x}{y^2} + \frac{\partial u}{\partial b} \times \frac{1}{z} \\
 &= -\frac{\partial u}{\partial a} \frac{x}{y^2} + \frac{\partial u}{\partial b} \times \frac{1}{z} \longrightarrow (2)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial u}{\partial z} &= \frac{\partial u}{\partial b} \times \frac{\partial b}{\partial z} + \frac{\partial u}{\partial c} \times \frac{\partial c}{\partial z} \\
 &= -\frac{y}{z^2} \frac{\partial u}{\partial b} + \frac{\partial u}{\partial c} \times \frac{1}{x} \longrightarrow (3)
 \end{aligned}$$

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} + z \cdot \frac{\partial u}{\partial z}$$

$$= 0$$

$$\text{ie, } \left[x \frac{\partial u}{\partial a} \times \frac{1}{y} - \frac{\partial u}{\partial c} \times \frac{z}{x} \right] + \left[-\frac{\partial u}{\partial a} \frac{x}{y} + \frac{\partial u}{\partial b} \frac{y}{z} \right]$$

$$+ \left[-\frac{y}{z} \frac{\partial u}{\partial b} + \frac{\partial u}{\partial c} \frac{z}{x} \right]$$

$$= \underline{\underline{0}}$$

b) Critical points

$$f_x = 2y - 3x^2 = 0 \quad \text{ie, } 2y - 3x^2 = 0 \longrightarrow (1)$$

$$f_y = 2x - 2y$$

$$2x - 2y = 0$$

$$x = y \rightarrow (2)$$

Sub. eqn (2) in (1)

$$2x - 3x^2 = 0$$

$$x(2 - 3x) = 0$$

$$\Rightarrow x = 0 \text{ or } 2 - 3x = 0$$

$$2 - 3x = 0$$

$$3x = 2$$

$$x = \frac{2}{3}$$

$\therefore$ Critical points are $(0,0)$ $(\frac{2}{3}, \frac{2}{3})$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

$$f_{xx} = -6x$$

$$f_{yy} = -2$$

$$f_{xy} = 2$$

at $(0,0)$

$$D = (-6 \times 0) \times (-2) - 2^2$$

$$= -4$$

$D < 0 \therefore (0,0)$ is a saddle point.

at $(\frac{2}{3}, \frac{2}{3})$

$$D = \left(-6 \times \frac{2}{3}\right) \times -2 - 2^2$$

$$= \underline{\underline{-4}}$$

$$D > 0$$

$$f_{xx} = -4 < 0$$

$\therefore p$ has relative maximum at $(\frac{2}{3}, \frac{2}{3})$

December 2020

MODULE-III

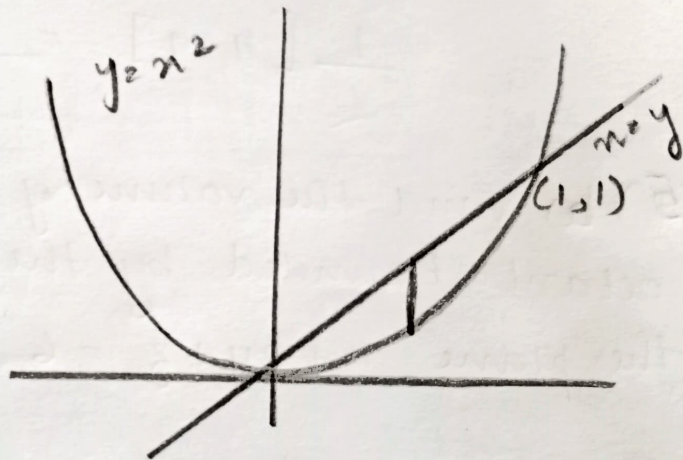
5) Find the mass of the lamina with density function $\rho(x, y) = x^2$ which is bounded by $y = x$ and $y = x^2$

ans: $\int (x, y) = x^2$

$$\text{mass, } M = \iint_R \rho(x, y) dA$$

$$y = x^2$$

$$y = x$$



$$y: x^2 \rightarrow x$$

$$x: 0 \rightarrow 1$$

$$M = \int_0^1 \int_{x^2}^x x^2 dy dx$$

$$\begin{cases} x = x^2 \\ x - x^2 = 0 \\ x(1-x) = 0 \\ x = 0 & x = 1 \\ y = 0 & y = 1 \end{cases}$$

$$= \int_0^1 n^2 [y]_{n^2}^n dn$$

$$= \int_0^1 n^2 [n - n^2] dn = \int_0^1 n^3 - n^4 dn$$

$$= \left[\frac{n^4}{4} - \frac{n^5}{5} \right]_0^1 = \left[\frac{1}{4} - \frac{1}{5} \right] - 0$$

$$= \frac{1}{20} //$$

6.) Evaluate $\iint_R y^2 n dA$ over the region $R = \{(n, y) - 3 \leq n \leq 2, 0 \leq y \leq 1\}$.

ans: $\iint_R y^2 n dA = \int_{-3}^2 \int_0^1 y^2 n dy dn$

$$= \int_{-3}^2 \left[\frac{y^3}{3} \right]_0^1 n dn = \frac{1}{3} \int_{-3}^2 n dn = \frac{1}{3} \left[\frac{n^2}{2} \right]_{-3}^2$$

$$= \frac{1}{6} [4 - 9] = \frac{1}{6} \times -5 = \frac{-5}{6} //$$

15 a) use double integral to find the area of the region enclosed b/w the parabola, $y = \frac{x^2}{2}$ and the line $y = 2x$.

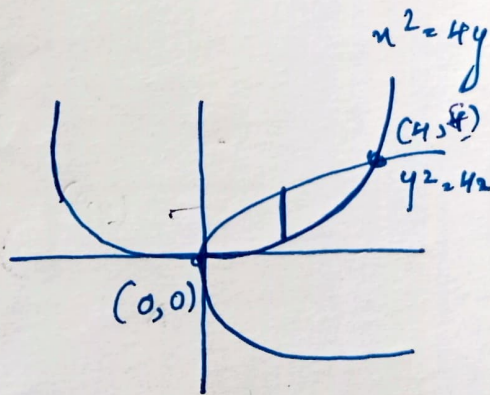
$$\text{ans: } \int_0^4 \int_{x^2/2}^{2x} dy dx = \int_0^4 [y]_{\frac{x^2}{2}}^{2x} dx$$

$$= \int_0^4 \left[2x - \frac{x^2}{2} \right] dx$$

$$= \left[\frac{2x^2}{2} - \frac{x^3}{6} \right]_0^4 = 4^2 - \frac{4^3}{6}$$

$$= 16 - \frac{32}{3}$$

$$= \frac{16}{3}$$



$$x^2 = 4y$$

$$x^2 = 4 \cdot 2\sqrt{x}$$

$$x^2 = 8\sqrt{x}$$

$$x^{3/2} = 8$$

$$x^{1/2} = 2$$

$$x = 4 \Rightarrow y = 8$$

$$y: \frac{x^2}{4} \rightarrow \sqrt{4}x$$

$$x: 0 \rightarrow 4$$

to find the area

15) b) Find the volume of the solid in the first octant bounded by the coordinate planes and the plane $x + 2y + z = 6$

ans: $x + 2y + z = 6$

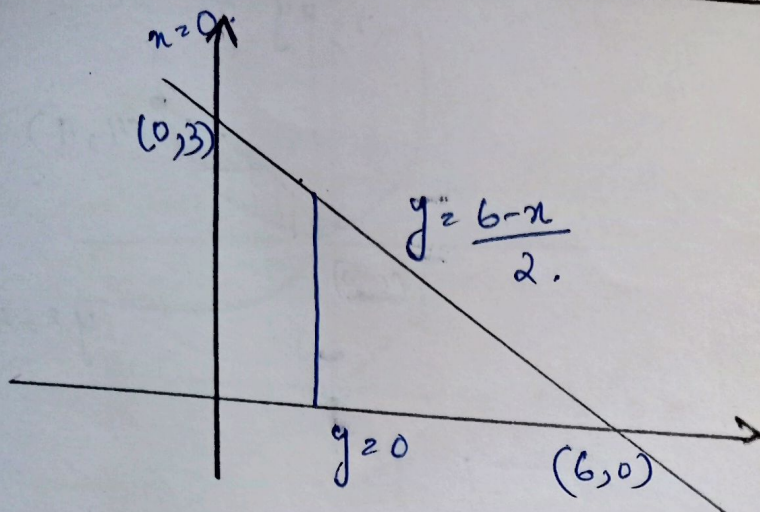
$$\therefore z = 6 - x - 2y$$

$$\therefore z = 0 \rightarrow 6 - x - 2y$$

put $z = 0$

$$x + 2y = 6$$

x	$y = 6 - x/2$
0	3
6	5/2



$$y: 0 \rightarrow \frac{6-x}{2}$$

$$x: 0 \rightarrow 6$$

$$\text{Vol} = \iiint_R dA = \int_0^6 \int_0^{\frac{6-x}{2}} \int_0^{6-x-2y} dz dy dx$$

$$\int_0^6 \int_0^{\frac{6-x}{2}} [z]_0^{6-x-2y} dy dx = \int_0^6 \int_0^{\frac{6-x}{2}} (6-x-2y) dy dx$$

$$= \int_0^6 \left[6y - \frac{2y^2}{2} \right]_0^{\frac{6-x}{2}} dx$$

$$= \int_0^6 \frac{1}{2} \left[36 - 6x - 6x + x^2 - \frac{36}{2} + 6x - \frac{x^2}{2} \right] dx$$

$$= \frac{1}{2} \int_0^6 \left[18 - 6x + \frac{x^2}{2} \right] dx = \frac{1}{2} \left[18x - \frac{6x^2}{2} + \frac{x^3}{6} \right]_0^6$$

$$= \frac{1}{2} \left[18 \times 6 - 3 \times 6^2 + \frac{6^3}{6} \right] = \frac{1}{2} [36] = 18 //$$

16) a) change the order of integration and hence evaluate $\int_0^4 \int_{x^2/4}^{2\sqrt{x}} dy dx$.

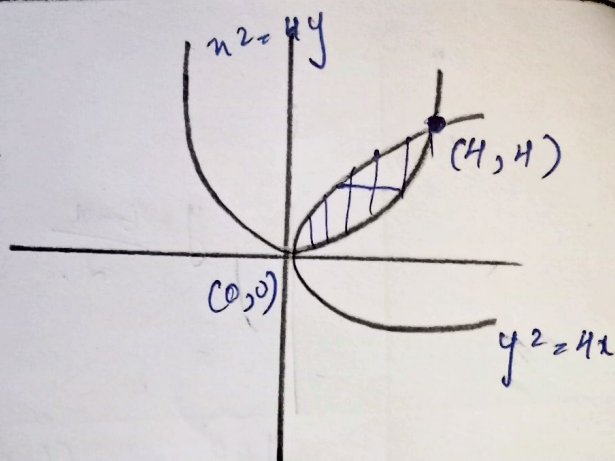
ans: $y: \frac{x^2}{4} \rightarrow 2\sqrt{x}$

$$y = \frac{x^2}{4} \Rightarrow x^2 = 4y //$$

$$y = 2\sqrt{x}$$

$$\frac{y}{2} = \sqrt{x} \Rightarrow x = \frac{y^2}{4}$$

$$y^2 = 4x$$



$$x: 0 \rightarrow 4$$

$$\frac{x^2}{4} = 2\sqrt{x}$$

$$x^2 = 8\sqrt{x} \Rightarrow x^4 = 64x$$

$$x^4 - 64x = 0 \Rightarrow x(x^3 - 64) = 0$$

$$x = 0 \text{ or } x^3 = 64 \therefore x = 4, y = 4 //$$

$$= \int_0^4 \int_{y^2/4}^{2\sqrt{y}} dx dy$$

$$\int_0^4 \left[x^2 \right]_{y^2/4}^{2\sqrt{y}} dy = \int_0^4 \left[2\sqrt{y} - y^2/4 \right] dy = \frac{1}{4} \int_0^4 8\sqrt{y} - y^2 dy$$

$$= \frac{1}{4} \left[8 \times \frac{y^{3/2}}{3/2} - \frac{y^3}{3} \right]_0^4 = \frac{1}{4} \left[\frac{16y^{3/2}}{3} - \frac{y^3}{3} \right]_0^4$$

$$= \frac{1}{12} \left[16 \times 2^3 - y^3 \right]_0^4 = \frac{1}{12} [16 \times 8 - 64]$$

$$= \frac{1}{12} \times 64 = \frac{16}{3} //$$

b) Evaluate $\iiint_C z dv$ where C is the in
the 1st octant cut off from the cylindrical
solid $y^2 + z^2 \leq 1$ and the planes $y = x$ and $x = 0$

ans: $y = x$ $y^2 + z^2 \leq 1$ $x = 0$

put $z = 0$

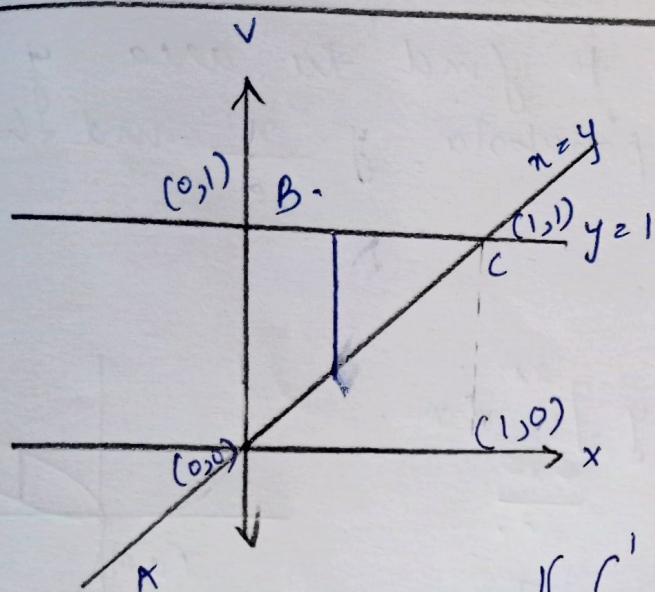
$y \leq 1$

$y \leq 1$

$z^2 \leq 1 - y^2$

$z \leq \sqrt{1 - y^2}$

$x: 0 \rightarrow \sqrt{1 - y^2}$



$$y: n \rightarrow 1$$

$$n: 0 \rightarrow 1$$

$$\int_0^1 \int_n^1 \int_0^{\sqrt{1-y^2}} z \, dz \, dy \, dx$$

ans: ~~given $x^2 + y^2 + z^2 \leq 1$~~

$$= \int_0^1 \int_n^1 \left[\frac{z^2}{2} \right]_0^{\sqrt{1-y^2}} dy \, dx$$

$$= \frac{1}{2} \int_0^1 \int_n^1 [1-y^2] dy \, dx = \frac{1}{2} \int_0^1 \left[y - \frac{y^3}{3} \right]_n^1 dx$$

$$= \frac{1}{2} \int_0^1 \left[1 - \frac{1}{3} \right] - \left[n - \frac{n^3}{3} \right] dx$$

$$= \frac{1}{2} \int_0^1 \left[\frac{2}{3} - n + \frac{n^3}{3} \right] dx$$

$$= \frac{1}{2} \left[\frac{2}{3} x - \frac{n^2}{2} + \frac{n^4}{12} \right]_0^1$$

$$= \frac{1}{2} \left[\frac{2}{3} - \frac{1}{2} + \frac{1}{12} \right]$$

$$= \frac{1}{2} \left[\frac{8-6+1}{12} \right] = \frac{3}{24}$$

$$= \frac{1}{8}$$

MODULE -IV

7. Test the convergence of the series $\sum_{k=1}^{\infty} \left(\frac{k}{100}\right)^k$

Soln:

$$a_k = \left(\frac{k}{100}\right)^k$$

$$\rho = \lim_{k \rightarrow \infty} a_k^{1/k} = \lim_{k \rightarrow \infty} \left(\left(\frac{k}{100}\right)^k\right)^{1/k} = \lim_{k \rightarrow \infty} \frac{k}{100}$$

$$= \infty$$

$\therefore$ series diverges by Root test.

8. Does the series $\sum_{k=1}^{\infty} \left(-\frac{3}{4}\right)^k$ converge? If so find its

Sum.

Soln.

It is a geometric series with $a = -\frac{3}{4}$

$$r = -\frac{3}{4}$$

$$|r| = \frac{3}{4} < 1$$

$\therefore$ series converges.

$$\text{Sum} = \frac{a}{1-r} = \frac{-3/4}{1 - (-3/4)} = \frac{-3/4}{1 + 3/4} = \frac{-3/4}{7/4}$$

$$= \underline{\underline{-3/7}}$$

17 a) Test the convergence of the series $1 + \frac{1 \cdot 3}{3!} + \frac{1 \cdot 3 \cdot 5}{5!} + \dots$

Soln:

$$a_k = \frac{1 \cdot 3 \cdot \dots \cdot (2k-1)}{(2k-1)!}$$

$$a_{k+1} = \frac{1 \cdot 3 \cdot \dots \cdot (2k-1) \cdot (2k+1)}{(2k+1)!}$$

$$\rho = \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \left[\frac{1 \cdot 3 \cdot \dots \cdot (2k-1) \cdot (2k+1)}{(2k+1)!} \cdot \frac{(2k-1)!}{1 \cdot 3 \cdot \dots \cdot (2k-1)} \right]$$

$$= \lim_{k \rightarrow \infty} \frac{(2k-1)! \cdot (2k+1)}{(2k-1)! \cdot 2k \cdot (2k+1)} = \lim_{k \rightarrow \infty} \frac{1}{2k}$$

$$= 0 < 1$$

$\therefore$ By Ratio test, the series converges.

17 b) Find the sum of the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$

Soln:

$$a_k = \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{(k+1)}$$

$$S_n = a_0 + a_1 + \dots + a_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right)$$

$$+ \dots + \left(\frac{1}{n} - \frac{1}{(n+1)}\right)$$

$$= 1 - \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1$$

$$\therefore \underline{\underline{\text{Sum} = 1}}$$

18 a) Test the convergence of 1) $\sum_{k=1}^{\infty} \frac{k!}{3!(k-1)! 3^k}$

2) $\sum_{k=1}^{\infty} \left(\frac{4k-5}{2k+1} \right)^k$

Soln:

$$a_k = \frac{k!}{3!(k-1)! 3^k}$$

$$a_{k+1} = \frac{(k+1)!}{3! k! 3^{k+1}}$$

$$\rho = \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{(k+1)!}{3! k! 3^{k+1}} \cdot \frac{3! (k-1)! 3^k}{k!}$$

$$= \lim_{k \rightarrow \infty} \frac{(k+1)}{3k} = \frac{1}{3} \lim_{k \rightarrow \infty} \frac{1+1/k}{1}$$

$$= \frac{1}{3} < 1$$

$\therefore$ series converges by Ratio test.

2) $a_k = \left(\frac{4k-5}{2k+1} \right)^k$

$$\rho = \lim_{k \rightarrow \infty} a_k^{1/k} = \lim_{k \rightarrow \infty} \left(\left(\frac{4k-5}{2k+1} \right)^k \right)^{1/k}$$

$$= \lim_{k \rightarrow \infty} \frac{4k-5}{2k+1} = \lim_{k \rightarrow \infty} \frac{4-5/k}{2+1/k} = \frac{4}{2} = 2 > 1$$

$\therefore$ series diverges by Root test.

18 b) Test the absolute or conditional convergence of-

$$\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^2}{(k^3+1)}$$

Soln:

$$a_k = \frac{k^2}{k^3+1}$$

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} \frac{k^2}{k^3+1} = \lim_{k \rightarrow \infty} \frac{1/k}{1+1/k^3} = 0$$

$$a_{k+1} = \frac{(k+1)^2}{(k+1)^3+1}$$

$$a_k > a_{k+1}$$

$\therefore$ By alternating series test, $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^2}{k^3+1}$ converges

Consider the series of absolute values

$$\text{i.e. } \sum_{k=1}^{\infty} \frac{k^2}{k^3+1} \quad \therefore a_k = \frac{k^2}{k^3+1}$$

$$\text{By informal principle } \sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{k^2}{k^3} = \sum_{k=1}^{\infty} \frac{1}{k},$$

harmonic series $\therefore$ diverges.

$$\begin{aligned} \rho &= \lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{\frac{k^2}{k^3+1}}{\frac{1}{k}} = \lim_{k \rightarrow \infty} \frac{k(k^2)}{k^3+1} \\ &= \lim_{k \rightarrow \infty} \frac{1}{1+1/k^3} = 1 > 0 \text{ and finite} \end{aligned}$$

By Limit comparison test, since $\sum \frac{1}{k}$ diverges

$$\sum_{k=1}^{\infty} \frac{k^2}{k^3+1} \text{ also diverges.}$$

$$\therefore \sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^2}{k^3+1} \text{ diverges absolutely.}$$

$$\therefore \sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^2}{k+1} \text{ conditionally convergent.}$$

PART A

9. Find the binomial series for $f(x) = (1+x)^{1/3}$ upto third degree term.

Binomial series for $f(x)$ is,

$$f(x) = (1+x)^m = 1 + mx + \frac{m(m-1)}{2!} x^2 + \dots$$

$$(1+x)^{1/3} = 1 + \frac{1}{3}x + \frac{\frac{1}{3}(\frac{1}{3}-1)}{2!} x^2 + \dots$$

$$= 1 + \frac{1}{3}x + \frac{\frac{1}{3} \times \frac{-2}{3}}{2} x^2 + \dots$$

$$= 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \dots$$

10. Find the Maclaurin's series of $f(x) = \log(1+x)$ upto third degree term.

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \dots$$

$$f(x) = \log(1+x)$$

$$f'(x) = \frac{1}{1+x}$$

$$f''(x) = \frac{-1}{(1+x)^2}$$

$$f'''(x) = \frac{2}{(1+x)^3}$$

$$f(0) = \log 1 = 0$$

$$f'(0) = 1$$

$$f''(0) = -1$$

$$f'''(0) = \frac{2}{1^3} = 2$$

$$\therefore f(x) = 0 + x \times 1 + \frac{x^2}{2!} \times (-1) + \frac{x^3}{3!} \times 2 + \dots$$

$$= x + \frac{x^2}{2!} - \frac{x^3}{3} + \dots$$

19 a) Expand into Fourier series, $f(x) = e^{-x}$, $0 < x < 2\pi$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{2l} \int_{-l}^l f(x) dx$$

$$2l = 2\pi$$

$$l = \pi$$

$$= \frac{1}{2\pi} \int_0^{2\pi} e^{-x} dx$$

$$= \frac{1}{2\pi} \left[\frac{e^{-x}}{-1} \right]_0^{2\pi} = \frac{1}{2\pi} [e^{-2\pi} - e^0]$$

$$= \frac{1}{2\pi} [e^{-2\pi} - 1] = \frac{1}{2\pi} [1 - e^{-2\pi}]$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cdot \cos(nx) dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{1+n^2} (1 \cos nx + n \sin nx) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \frac{e^{-2\pi}}{1+n^2} [\cos n2\pi + 2\pi \sin n2\pi] - \left(\frac{e^0}{1+n^2} (\cos 0 + n \sin 0) \right) \right\}$$

$$= \frac{1}{\pi} \left\{ \frac{e^{-2\pi}}{1+n^2} [\cos 2n\pi] - \frac{1}{1+n^2} [1] \right\}$$

$$= \frac{1}{\pi(1+n^2)} [1 - e^{-2\pi}]$$

$$b_m = \frac{1}{\pi} \int_0^{2\pi} e^{-x} \sin(nx) dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{1+m^2} (-\sin nx - n \cos nx) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\frac{e^{-2\pi}}{1+m^2} (-n) - \frac{1}{1+m^2} (-n) \right]$$

$$= \frac{n}{\pi(m^2+1)} [1 - e^{-2\pi}]$$

$$\therefore f(x) = \frac{1 - e^{-2\pi}}{2\pi} + \sum_{n=1}^{\infty} \frac{1}{\pi(m^2+1)} (1 - e^{-2\pi}) \cos nx + \sum_{n=1}^{\infty} \frac{n}{\pi(m^2+1)} (1 - e^{-2\pi}) \sin nx$$

b) Find the half range cosine series for $f(x) = (x-1)^2$ in $0 \leq x \leq 1$

Half range cosine series of $f(x)$ is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l} \int_0^l f(x) dx$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$a_0 = \frac{1}{1} \int_0^1 (x-1)^2 dx$$

$$= \left[\frac{(x-1)^3}{3} \right]_0^1 = \frac{0}{3} - \frac{(-1)^3}{3} = \underline{\underline{\frac{1}{3}}}$$

$$a_n = \frac{2}{1} \int_0^1 (x-1)^2 \cos(n\pi x) dx$$

$$= 2 \left\{ (x-1)^2 \times \frac{\sin(n\pi x)}{n\pi} - 2(x-1) \cdot \frac{1}{n\pi} \times \frac{\cos(n\pi x)}{n\pi} + 2 \times \frac{1}{n^2 \pi^2} \times \frac{\sin(n\pi x)}{n\pi} \right\} \Bigg|_0^1$$

$$= 2 \left\{ 0 - \left[\frac{2}{n^2 \pi^2} \times -1 \times \cos 0 \right] \right\}$$

$$= \frac{2 \times 1}{n^2 \pi^2}$$

$$= \frac{4}{n^2 \pi^2}$$

$$f(x) = \frac{2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos(n\pi x)$$

20 a) Find the fourier series of the function $f(x) = |x|$ in $-1 \leq x \leq 1$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{2l} \int_{-l}^l f(x) dx$$

$$= \frac{1}{2} \int_{-1}^1 |x| dx = \frac{1}{2} \times 2 \int_0^1 x dx$$

$$= \left[x \times \frac{x^2}{2} \right]_0^1 = \underline{\underline{1/2}}$$

$$a_n = \frac{1}{1} \int_{-1}^1 |x| \cdot \cos(n\pi x) dx$$

$$= 2 \int_0^1 x \cos n\pi x dx$$

$$= 2 \left[x \times \frac{\sin(n\pi x)}{n\pi} - 1 \times \frac{1}{n\pi} \times \frac{-\cos(n\pi x)}{n\pi} \right]_0^1$$

$$= 2 \left[\frac{1}{n\pi} \sin(n\pi) + \frac{1}{n^2\pi^2} \cos(n\pi) - \left(0 + \frac{1}{n^2\pi^2} \cos 0 \right) \right]$$

$$= 2 \left[\frac{1}{n^2\pi^2} (-1)^n - \frac{1}{n^2\pi^2} \right]$$

$$= \frac{2}{n^2\pi^2} [(-1)^n - 1]$$

$$b_n = \frac{1}{1} \int_{-1}^1 |x| \sin(n\pi x) dx$$

$$\left(\because \int_{-a}^a f(x) dx = 0 \right. \\ \left. \text{if } f(x) \text{ is odd} \right)$$

$$= 0$$

$$\therefore f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2} ((-1)^n - 1) \cos(n\pi x)$$

20 b) Find the Fourier sine series of $f(x) = x \cos x$ in $0 < x < \pi$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \cos x \sin(n\pi x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \left[\frac{\sin(n+1)x + \sin(n-1)x}{2} \right] dx$$

$$\boxed{\begin{aligned} \sin A \cos B \\ = \frac{\sin(A+B) + \sin(A-B)}{2} \end{aligned}}$$

$$= \frac{1}{\pi} \left[\int_0^{\pi} x [\sin(n+1)x + \sin(n-1)x] dx \right]$$

$$= \frac{1}{\pi} \left[\int_0^{\pi} x \sin(n+1)x dx + \int_0^{\pi} x \sin(n-1)x dx \right]$$

$$= \frac{1}{\pi} \left[\left[x \times \frac{-\cos(n+1)x}{n+1} - 1 \times \frac{-1}{n+1} \cdot \frac{\sin(n+1)x}{n+1} \right]_0^{\pi} \right.$$

$$\left. + \left[x \times \frac{-\cos(n-1)x}{n-1} - 1 \times \frac{-1}{n-1} \cdot \frac{\sin(n-1)x}{n-1} \right]_0^{\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ \pi \times \frac{-1}{n+1} \cos(n+1)\pi + 0 + \pi \times \frac{-1}{n-1} \cos(n-1)\pi \right\}$$

$$= \frac{1}{\pi} \times \pi \left[\frac{-1}{n+1} (-1)^{n+1} - \frac{1}{n-1} (-1)^{n-1} \right]$$

$$= (-1)^{n-1} \left[\frac{-1}{n+1} - \frac{1}{n-1} \right] = (-1)^n \left[\frac{1}{n+1} + \frac{1}{n-1} \right]$$

$$= \frac{2n}{n^2-1} (-1)^n, n \neq 1$$

When $n=1$, $b_1 = \frac{2}{\pi} \int_0^{\pi} x \sin x \cos x dx$

$$= \frac{2}{\pi} \int_0^{\pi} \frac{x}{2} \sin 2x dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \sin 2x dx$$

$$= \frac{1}{\pi} \left[x \times \frac{-\cos 2x}{2} - 1 \times \frac{-1}{2} \times \frac{\sin 2x}{2} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\pi}{2} \times \cos 2\pi \right] - 0$$

$$= \frac{1}{\pi} \times \frac{\pi}{2} \times 1$$

$$= \frac{1}{2}$$

$$\therefore f(x) = \frac{-1}{2} \sin x + \sum_{n=2}^{\infty} \frac{2n}{n^2-1} (-1)^n \sin nx$$